Formalizing Fefferman's Official Periodic Navier–Stokes Problem (B) in Lean 4: Global Linear Strain Exclusion on $\mathbb{T}^3$, Biot–Savart Direction Depletion Identities, and Conditional Reductions via Beale–Kato–Majda

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Abstract

We present a self-contained Lean 4 formalization (`leanprover/lean4:v4.30.0-rc1`, Mathlib 4, 208 kernel-verified theorems across 6 modules with zero `sorry` and zero non-standard axioms outside `[propext, Classical.choice, Quot.sound]`) of Charles L. Fefferman's official periodic 3D incompressible Navier–Stokes problem statement (*Existence and Smoothness of the Navier–Stokes Equation*, Clay Mathematics Institute, 2006, Statement (B) on $\mathbb{R}^3/\mathbb{Z}^3$) together with unconditional geometric/kinematic identities, spatial-maximum Bochner–polar reduction, and explicit **conditional reductions** to Beale–Kato–Majda continuation. In `FeffermanClayNS/FinalProductTheorem.lean`, all five unconditionally proved components are packaged into the single 5-conjunct master product theorem `fefferman_periodic_ns_exact_reduction_product_theorem` $(Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu))$.

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1 Introduction and Single-Sentence Final Product Form $(Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu))$

In his official Clay Mathematics Institute problem description [1], Charles L. Fefferman formulates the periodic existence and smoothness problem (Statement (B)) for the three-dimensional incompressible Navier–Stokes equations on $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. For the unmodified Navier–Stokes PDE ($\kappa = 0$ in the dynamics), Lean's kernel cannot derive $\langle \omega, (\omega\cdot\nabla)u\rangle + \nu\langle \omega, \Delta\omega\rangle \leq C_{\mathrm{CZ}}M^3 - \kappa'' M^4$ at a spatial maximum without either taking the high-set direction-curvature bound $|\nabla\hat{\xi}(x_{\max})|^2 \geq c_{\mathrm{dir}}M$ as an explicit hypothesis or proving that open PDE estimate from scratch. Accordingly, this repository states **only the theorems it proves explicitly and unconditionally**, packaged in `FeffermanClayNS/FinalProductTheorem.lean` into a single 5-conjunct product theorem.

1.1 Single-Sentence English Prose Statement

Theorem 1.1 (Master Product Theorem: `fefferman_periodic_ns_exact_reduction_product_theorem`). *For every viscosity $\nu \in \mathbb{R}$ on the 3-torus $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, **($Q_1$)** spatial $\mathbb{Z}^3$-periodicity is equivalent to invariance under the three unit coordinate shifts $f(x + e_j) = f(x)$, and Fefferman's energy-bounded periodic statement implies his exact periodic statement (*$\mathrm{Clay\_B\_exact\_with\_energy}(\nu) \implies \mathrm{Clay\_B\_exact}(\nu)$*); **($Q_2$)** every $\mathbb{Z}^3$-periodic linear map $L : \mathbb{R}^3 \to \mathbb{R}^3$ vanishes identically ($L = 0$), excluding any non-zero global linear Burgers background strain* $\mathrm{diag}(-\alpha/2, -\alpha/2, \alpha)x$ *($\alpha \neq 0$) on $\mathbb{T}^3$; **(Q_3)** the Biot–Savart vortex-stretching determinant obeys* $\det(\omega(y), z, \omega(x)) = \det(\omega(y) - \omega(x), z, \omega(x))$ *and vanishes under collinear alignment $\omega(y) = c\,\omega(x)$, while under parabolic rescaling $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$ both cubic stretching $C_{\mathrm{CZ}}|\omega_\lambda|^3$ and viscous direction-curvature dissipation $\nu|\omega_\lambda|^2|\nabla\hat{\xi}_\lambda|^2$ scale homogeneously as λ^6; **(Q_4)** at a spatial maximum $x_{\max}$ where $\langle \omega, (\omega \cdot \nabla)u \rangle \leq C_{\mathrm{CZ}}M^3$, $\nu\langle \omega, \Delta\omega \rangle \leq -\nu M^2|\nabla\hat{\xi}|^2$, and high-set direction-curvature coercivity $|\nabla\hat{\xi}(x_{\max})|^2 \geq c_{\mathrm{dir}}M$ holds, the maximum vorticity rate satisfies $M'(t) \leq C_{\mathrm{CZ}}M(t)^2 - (\nu c_{\mathrm{dir}})M(t)^3$, which forces the uniform Lyapunov ceiling $M(t) \leq \max(M(0), C_{\mathrm{CZ}}/(\nu c_{\mathrm{dir}}))$; and **($Q_5$)** whenever a periodic flow class satisfies the high-set direction-curvature depletion hypothesis* $\mathrm{PeriodicDirectionDepletionBound}(\nu)$ *and classical periodic Beale–Kato–Majda continuation* $\mathrm{PeriodicBKMContinuation}(\nu)$*, Fefferman's periodic statement* $\mathrm{Clay\_B\_exact}(\nu)$ *holds.*

1.2 Exact Mathematical Product Form $(Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu))$

$$
\begin{aligned}
\mathcal{Q}_{\mathrm{ClayB}}(\nu) \;\;:\Longleftrightarrow\;\; & \underbrace{\Big((\forall f : \mathbb{R}^3 \to \mathbb{R}^3,\ \mathrm{IsZ3Periodic}(f) \iff \forall x,\, j \in \{1,2,3\},\ f(x + e_j) = f(x)) \;\wedge\; (\mathrm{Clay\_B\_exa}}_{Q_1(\nu)\text{: Exact Fefferman Periodic Unit-Shift Equivalence \& Energy I}} \\
& \wedge\ \underbrace{\Big((\forall L : \mathbb{R}^3 \to_{\mathbb{R}} \mathbb{R}^3,\ \mathrm{IsZ3Periodic}(L) \implies L = 0\Big) \;\wedge\; \Big(\forall \alpha \neq 0,\ \neg\mathrm{IsZ3Periodic}(\mathrm{diag}(-\tfrac{\alpha}{2}, -\tfrac{\alpha}{2}}_{Q_2\text{: Torus Exclusion of Global Linear Burgers Strain}} \\
& \wedge\ \underbrace{\Big(\det(\omega_y, z, \omega_x) = \det(\omega_y - \omega_x, z, \omega_x) \;\wedge\; \det(c\omega_x, z, \omega_x) = 0 \;\wedge\; C_{\mathrm{CZ}}(\lambda^2|\omega|)^3 - \nu(\lambda^2|\omega|)^2(\lambda|}_{Q_3\text{: Biot–Savart Alignment Depletion \& } \lambda^6 \text{ Parabolic Scaling Homogene}} \\
& \wedge\ \underbrace{\Big((\mathrm{Stretch} \leq C_{\mathrm{CZ}}M^3 \wedge \mathrm{Diff} \leq -\nu M^2|\nabla\hat{\xi}|^2 \wedge |\nabla\hat{\xi}|^2 \geq c_{\mathrm{dir}}M \implies M' \leq C_{\mathrm{CZ}}M^2 - \nu c_{\mathrm{dir}}M^3}_{Q_4\text{: Spatial-Maximum Bochner–Polar Reduction \& C}} \\
& \wedge\ \underbrace{\Big((\mathrm{PeriodicDirectionDepletionBound}(\nu) \;\wedge\; \mathrm{PeriodicBKMContinuation}(\nu)) \implies \mathrm{Clay\_B\_}}_{Q_5(\nu)\text{: Exact Conditional Implication to Fefferman's Clay\_B\_exact}(\nu)}
\end{aligned}
\tag{1}
$$

2 Verbatim Lean 4 Kernel Audit and Cryptographic Attestation

Compiling the repository with `lake build && lake env lean AxiomAudit.lean` executes an in-kernel traversal of the Lean 4 environment via `Lean.collectAxioms` across all 208 theorems in the 6 modules:

Listing 1: Verbatim output of `lake env lean AxiomAudit.lean` on `fefferman-clay-navier-stokes`.

```
FeffermanClayNS.TorusStrainExclusion: theorems=19 nonstandard=0
FeffermanClayNS.HighSetDepletionAndBKM: theorems=9 nonstandard=0
```

Table 1: Summary of Unconditionally Proved Theorems vs. Open Analytic Hypotheses in `FeffermanClayNS` (208 Theorems, 0 `sorry`).

Declaration / Mathematical Item	**Status**	**Mathematical Meaning & Scope**
`fefferman_periodic_ns_exact_reduction_product_theorem` (`FinalProductTheorem.lean`)	**PROVED** (Unconditional, 24 theorems)	Master 5-conjunct product theorem $(Q_1(\nu) \wedge Q_2 \wedge Q_3 \wedge Q_4 \wedge Q_5(\nu))$ and spatial-maximum Bochner–polar reduction.
`Clay_B_exact` ν, `isZ3Periodic_iff_unit_shifts`, `clay_B_exact_of_with_energy`	**PROVED** (Unconditional)	Exact encoding of Fefferman's Statement (B), unit-shift periodicity equivalence, and logical implication from the energy-bounded variant.
`z3_periodic_linear_eq_zero`, `burgersLinearStrain_not_z3_periodic`	**PROVED** (Unconditional)	Any $\mathbb{Z}^3$-periodic linear map on $\mathbb{R}^3$ vanishes identically; the global linear Burgers strain $\mathrm{diag}(-\alpha/2, -\alpha/2, \alpha)x$ ($\alpha \neq 0$) is not $\mathbb{Z}^3$-periodic.
`biot_savart_det_aligned_eq_zero`, `aligned_vorticity_stretching_integrand_zero`, `biot_savart_det_difference_identity`	**PROVED** (Unconditional)	Exact pointwise vanishing of $\det(\omega(y), z, \omega(x))$ when $\omega(y) = c\,\omega(x)$, reducing the stretching kernel to direction increments $\omega(y) - \omega(x)$.
`first_remaining_gap_bochner_polar_damping_at_max`, `scale_invariant_cubic_rate_of_direction_coercivity`, `comparison_ode_ceiling`	**PROVED** (Unconditional)	Spatial-maximum Bochner–polar reduction from $\lvert\nabla\hat{\xi}\rvert^2 \geq c_{\mathrm{dir}}\lvert\omega\rvert$ to $M' \leq C_{\mathrm{CZ}}M^2 - \kappa'' M^3$ and ODE comparison ceiling $M(t) \leq \max(M(0), C_{\mathrm{CZ}}/\kappa'')$.
`clay_B_exact_of_periodic_tether_bound`, `clay_B_exact_of_torus_direction_depletion`	**PROVED CONDITIONAL REDUCTIONS**	Conditional theorems proving that *if* the open direction-depletion/tether hypothesis and BKM continuation hold, then `Clay_B_exact` ν follows.
`periodicFiniteTimeVorticityBound_of_uniform`, `periodicBKMContinuation_of_finiteTime`, `clay_B_exact_of_periodic_finite_time_bound`	**PROVED CONDITIONAL REDUCTIONS**	Statement (B) needs only, for each horizon T, a bound $\lVert\omega(t)\rVert_\infty \leq C(T)$ on $[0, T]$ (`PeriodicFiniteTimeVorticityBound`, OPEN), not a bound uniform in time. Uniform ceiling ⇒ finite-horizon bound; finite-horizon BKM continuation ⇒ `PeriodicBKMContinuation`; `PeriodicFiniteTimeVorticityBound` ∧ `PeriodicBKMContinuationFiniteTime` ⇒ `Clay_B_exact`.

```
FeffermanClayNS.FeffermanStatements: theorems=13 nonstandard=0
FeffermanClayNS.Defs: theorems=143 nonstandard=0
FeffermanClayNS.FinalProductTheorem: theorems=24 nonstandard=0
TOTAL theorems=208 nonstandard=0
'FeffermanClayNS.fefferman_periodic_ns_exact_reduction_product_theorem' depends on axioms: [
    propext, Classical.choice, Quot.sound]
```

References

[1] C. L. Fefferman, *Existence and Smoothness of the Navier–Stokes Equation*, in *The Millennium Prize Problems*, Clay Mathematics Institute / American Mathematical Society, pp. 57–67 (2006).

[2] J. T. Beale, T. Kato, and A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Commun. Math. Phys. **94** (1984), 61–66.

[3] P. Constantin and C. Fefferman, *Direction of vorticity and the problem of global regularity for the Navier–Stokes equations*, Indiana Univ. Math. J. **42** (1993), 775–789.

[4] T. Kato, *Nonstationary flows of viscous and ideal fluids in* $\mathbb{R}^3$, J. Funct. Anal. **9** (1972), 296–305.

[5] B. S. Frohman, *Titan Audit: Specification Gaming and Semantic Vacuity Detection for Formal Verification in Lean 4*, Zenodo, DOI: 10.5281/zenodo.23630648 (2026).

[6] B. S. Frohman, *SealGuard: Cryptographic Containment, Capability-Attested Boundary Enforcement, and Runtime Integrity Architecture*, Zenodo, DOI: 10.5281/zenodo.23631670 (2026).